# Autonomous mapping robot – project plan

**The plan might change during the course of the project**

**1st week – Ordering components (19.10.2026 – 25.10.2026)**

We will begin the project by purchasing components, which includes ordering the necessary hardware, such as an ESP32-S3 microcontroller, motors with encoder, power supply system components, and other required items. The list of items may change during the assembly process.

**2nd week – Mechanical construction and power supply (26.10.2026 – 1.11.2026)**

Gathering the components. Assembling robot chassis. Installing the JGA25-370 motors with encoders. Setting up the power supply system: a holder for two 18650 cells, adding a step-down converter to reduce the voltage. Printing the chassis on 3D printer.

Milestone: A rigid, assembled platform. Optimal placement of basic components and working power supply.

**3rd week – Drive implementation and ESP32-S3 environment (2.11.2026 – 8.11.2026)**

Connecting the pins of our motor driver to the ESP32. Implementing basic movement functions: moving forward and backwards, turning left and right.

Milestone: Testing motion of our robot, if it's capable of rotating etc.

**4th week – Odometry and interrupts implementation (9.11.2026 – 15.11.2026)**

Connecting the encoders of DC motors to GPIO pins. Writing hardware interrupts. Implementing math and calculating the robot's global coordinates and its orientation angle.

Milestone: Getting correct coordinates of our robot, after test drive.

**5th week – Construction of our own LIDAR system (16.11.2026 – 22.11.2026)**

Building a custom LIDAR system using a laser distance sensor and servomechanism.

Milestone: Obtaining accurate distance and angle data from a LIDAR system.

**6th week – LIDAR and encoder synchronization (23.11.2026 – 29.11.2026)**

Combining the data from LIDAR system with readings from rotary encoders to achieve precise localization.

Milestone: Complete, dynamically updated data table containing outlines of obstacles.

**7th week – Implementing live visualization and camera on WWW server (30.11.2026 – 6.12.2026)**

**8th and 9th week – Final testing, time for corrections (7.12.2026 – 20.12.2026)**

Constructing the body, making any necessary adjustments and ensuring the safety and reliability of our autonomous robot.

Milestone: Fully working autonomous robot, capable of indoor mapping.